seeed studio

INDUSTRIAL PRODUCT DATASHEET

[10%off] reBot Arm B601-DM Bundle with Jetson Orin

The [10%off] reBot Arm B601-DM Bundle with Jetson Orin is a plug-and-play, fully open-source 6+1 DoF robotic arm for embodied AI applications including visual grasping, teleoperation, and precision tasks.

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Overview



The [10%off] reBot Arm B601-DM Bundle with Jetson Orin is a plug-and-play, fully open-source 6+1 DoF robotic arm kit paired with an NVIDIA Jetson Orin Nano 8G pre-installed with Jetpack. It supports ROS1/2, Isaac Sim, and LeRobot, and enables applications such as visual grasping, teleoperation, assistive automation, and precision tasks in constrained environments.

Key Features

- **Plug-and-play bundle:** Assembled robot arm kit plus NVIDIA Jetson Orin Nano 8G box pre-installed with Jetpack, allowing plug-and-play deployment.
- **Fully Open-Source:** A truly 100% open-source 6+1 DoF robotic arm in terms of hardware and software. It includes BOM lists, hardware drawings, software, and algorithms—everything you need.
- **High-performance motors**: 4 Damiao 4310 motors + 3 Damiao4340P high-torque, low-noise motors ensure stable and precise power for the reBot Arm B601-DM.
- **Framework Compatibility:** Supports mainstream robotics development frameworks such as ROS1/2, Isaac Sim, and LeRobot.
- **Constant Content Updates:** As the project progresses, we will gradually update the latest algorithms and launch a series of completely free courses.

⚠️Buying Guide

This is a **fully assembled version**, ready to use out of the box. **Items pre-selected are essential** for smooth robotic arm operation. If you'd like to **explore teleoperation**, please purchase the "Star Arm 102 (Leader Arm)" and "12V 2A AC/DC Adapter (5-in-1)" together. **A mounting G-clamp is required** to securely fix the robotic arm to a workbench. Please make sure to include one if you do not already have it. All other accessories are optional and can be purchased as needed.

Product Gallery













Specification

	reBot Arm B601-DM Assembled Kit with Gripper
Degrees of Freedom	6+1

Reach	767mm 607mm(without gripper)
Payload	1.5kg (without gripper, 70% reach recommended)
Joint Range of Motion	J1:±150° J2:-220 - 0° J3:-220 - 0° J4:±90° J5:±90° J6:±90° Gripper:-325° - 0°
Repeatability	0.2 mm
Weight	about 4.7kg
Servos	DM-J4340P-2EC x3 DM-J4310-2EC x4
Operating Temperature	-20°~50°
Power	DC 24V 10A
ROS1	✅
Moveit1	✅
ROS2	✅
Moveit2	✅
Python	✅
Lerobot	✅
Isaac Sim	✅
Pinocchio	✅
Open Source	✅

Hardware Overview

Application / Devices	Sim-to-Real VLA DLI Course	GraspNet	Voice-control
reBot B601-DM (plus power supply)		✓	✓
reBot B601-RS (plus power supply)	✓		
star 102 Arm (plus power supply)	✓		
NVIDIA Jetson AGX Thor Developer Kit	✓		✓
NVIDIA Jetson Orin Super Developer Kit		✓	
2D Camera (Could use any ET-S231 USB camera from vision accessories list)	✓		
3D Camera (Could use any Orbbec/Realsense camera from vision accessories list)		✓	✓
reSpeaker			✓

Applications

- **GraspNet Visual Grasping on Jetson:** Powered by the NVIDIA Jetson AI edge computing platform and reBot Arm B601-DM, this embodied AI vision-based grasping solution enables intelligent robotic interaction with the physical world. By integrating RGB-D depth cameras (Orbbec Gemini 2), it delivers accurate and autonomous 3D perception and robotic grasping. The system integrates advanced technologies including YOLO-based object detection, 3D point cloud processing, the GraspNet-Baseline grasping algorithm, and hand-eye calibration, making embodied AI development more accessible for researchers and developers.
- **MotorBridge control:** Seeed offers Motorbridge, an integrated joint control and robotic arm simulation interface. It delivers intuitive control capabilities and effectively narrows the sim-to-real gap.
- **Teleoperation & Remote Robotics Control:** The reBot Arm B601 enables intuitive teleoperation for remote manipulation, data collection, and human-in-the-loop training. Combined with AI robotics frameworks like Hugging Face LeRobot, it supports datasets and pre-trained model and policies, and imitation learning workflows that translate human actions into robotic behaviors, accelerating embodied AI development.
- **Assistive Automation for Home Daily Tasks:** The reBot Arm B601 supports the development of assistive robotics for everyday tasks, from pick-and-place to simple household automation. By combining vision and learning-based control, the robot can adapt to unstructured environments and variable objects, helping bring embodied AI into practical, human-centered scenarios.
- **Precision Tasks in Harsh or Constrained Environments:** With ±0.2 mm repeatability and stable actuation, the B601 is suited for precise tasks in challenging or unsafe environments. Developers can prototype applications such as inspection, sample handling, or fine manipulation under constrained conditions. Its support for dynamic motion execution enables more reliable interaction with uncertain or sensitive objects. This makes it a strong platform for validating robotics solutions before scaling to industrial systems.

Part List

Name	Quantity
Assembled Robotic Arm (with gripper)	1
Hex key wrench set	1
Mouse pad	1

Resource Links

- https://www.seeedstudio.com/reBot-Arm-B601-DM-Bundle-with-Jetson-Orin.html
- reBot Arm Github
- Tool
- reComputer J3011 Datasheet
- reComputer J3011 schematic
- reComputer J3011 3D File

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